HU5175 Education, Technology, and Society

Assignment 3A
Bloom's Taxonomy

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Introduction

According to Wikipedia,

"Taxonomy is the practice and science of categorization or classification.

"A taxonomy (or taxonomical classification) is a scheme of classification, especially a hierarchical classification, in which things are organized into groups or types... Many taxonomies are hierarchies (and thus, have an intrinsic tree structure), but not all are."

Bloom's taxonomy was devised by a committee of educators at the University of Chicago in the year 1956, which was chaired by Benjamin Bloom, an American educational psychologist, after whom this taxonomy was named. The other noteworthy collaborators were Max Englehart, Edward Furst, Walter Hill, and David Krathwohl.

It is a system of three hierarchical models used to categorise learning objectives into varying levels of complexity and specificity. It was primarily developed to provide a common language for teachers to discuss and exchange learning and assessment methods.

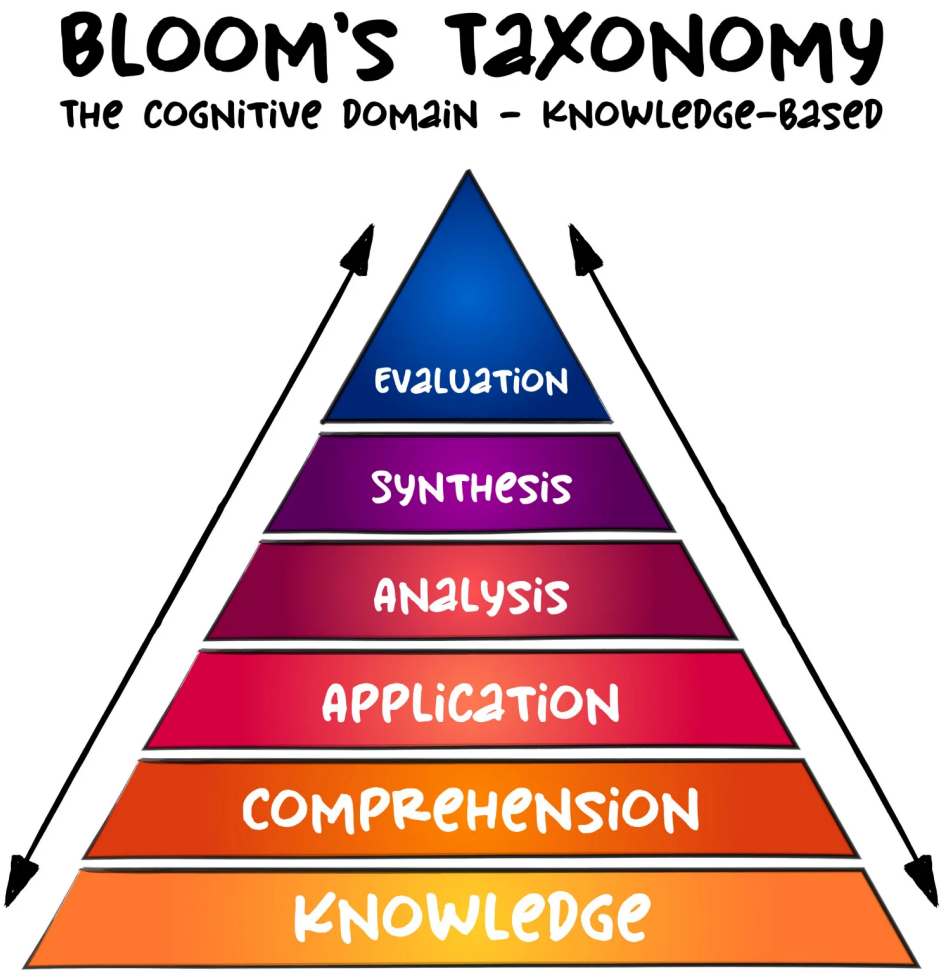
The three hierarchical models cover the learning objectives in the following domains:

- **Cognitive** - knowledge-based
- **Affective** - emotion-based
- **Psychomotor** - action-based

Bloom's taxonomy paved the way to align education goals, curricula, and examinations that are used in schools. It structures the breadth and depth of the activities and modules that teachers provided for students, and has had a ***profound impact on modern educational practice.***

While all the three domains of the taxonomy are crucial, it is the cognitive domain that has been the primary focus of most traditional education, and it is frequently used to structure curriculum learning objectives, assessments, and activities.

Cognitive Domain



In the original version of the taxonomy published in 1956, the cognitive domain is the first and most common of the learning objectives. It focuses on acquiring and applying knowledge, and is broken into six levels of objectives: knowledge, comprehension, application, analysis, synthesis, and evaluation.

The above diagram provides the hierarchical structure of the cognitive domain. It follows a bottom-up approach, meaning the higher levels are depended on having achieved the skills of the lower levels.

Knowledge	Recalling information, remembering facts
Comprehension	Making sense out of information, demonstrating an understanding of facts
Application	Using knowledge in a new but similar form, applying acquired knowledge
Analysis	Taking knowledge apart and exploring relationships, making inferences, generalising with evidence
Synthesis	Using information to create something new, bringing pieces of information together to form a new meaning
Evaluation	Examining relevant and available information to make judgements, judging the validity of ideas and the quality of work based on a set of criteria

With time, the cognitive model of Bloom's taxonomy evolved, and changes were brought forth to it that respected the foundation of the classification and yet adapted it to current times. Researchers had begun to rethink the way in which the educational objectives were presented by teachers, and they developed a revision of Bloom's taxonomy in 2001.

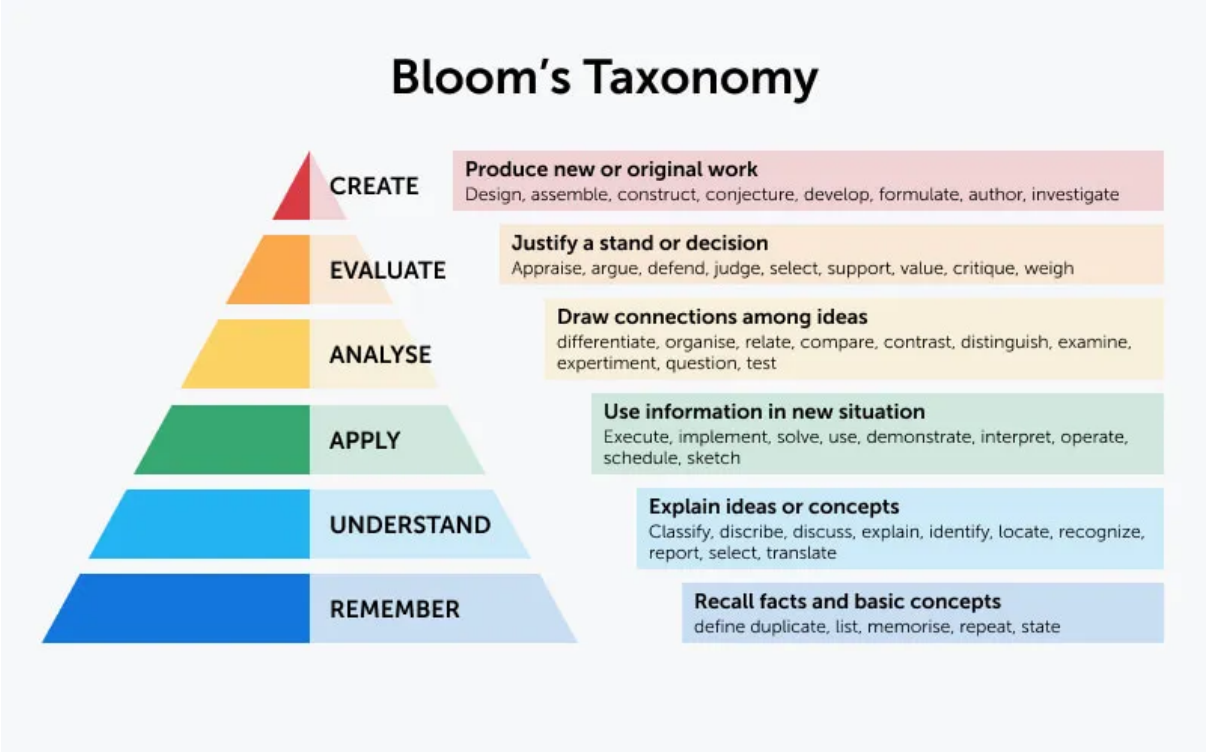
The revised taxonomy was developed by using many of the same processes and approaches that Bloom had used a half century earlier. In the new taxonomy, two dimensions are presented: the **knowledge** dimension and the **cognitive** dimension.

Knowledge is perhaps the most intuitive block of the cognitive model, but it was further broken down into four different types:

1. **Factual Knowledge** refers to knowledge of terminology and specific details.
2. **Conceptual Knowledge** describes knowledge of categories, principles, theories, and structures.
3. **Procedural Knowledge** encompasses all forms of knowledge related to specific skills, algorithms, techniques, and methods.
4. **Metacognitive Knowledge** defines knowledge related to thinking - knowledge about cognitive tasks and self-knowledge.

It is important to outline these different forms of knowledge to show how it is more dynamic than one may think and that there are multiple different types of knowledge that can be recalled before moving onto the next phase.

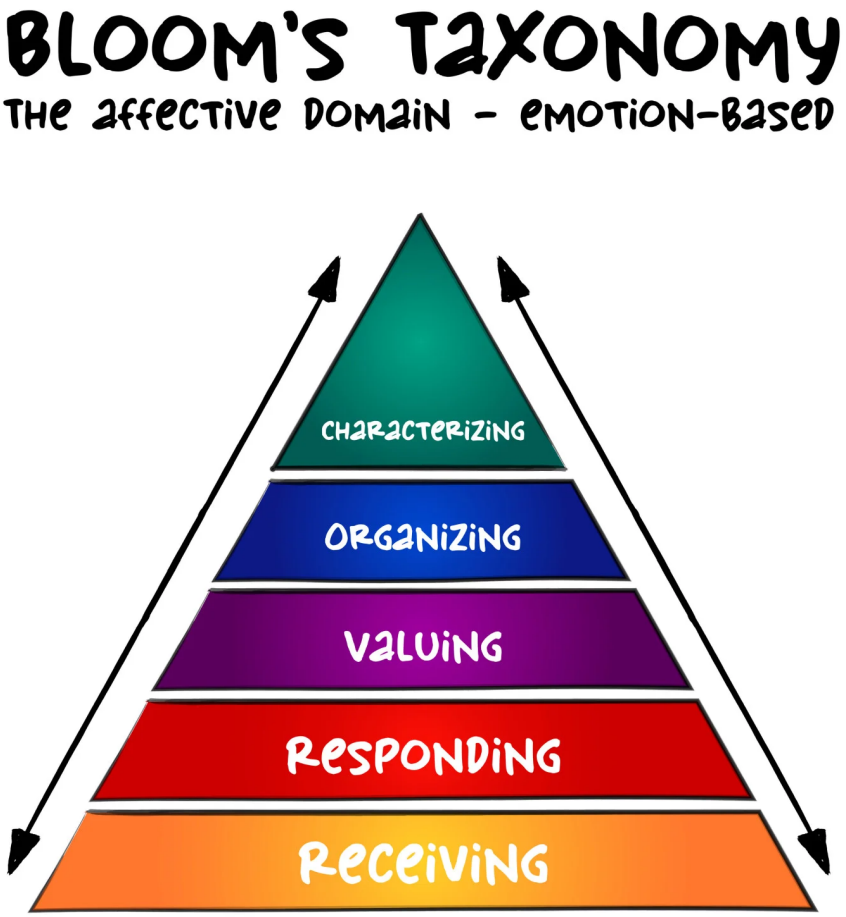
The cognitive process dimension makes minor changes to the original cognitive domain. The six levels now present are **remembering, understanding, applying, analysing, evaluating,** and **creating.**



The new taxonomy enabled teachers to think more in depth about the content that they are teaching and the objectives they are focusing on within the classroom. It allowed teachers to categorise objectives in a more-multidimensional way and to do so in a manner that allows them to see the complex relationships between knowledge and cognitive processes.

The original approach provided a structure for how people thought about facts, concepts, and generalisations and offered a common language for thinking about and communicating educational objectives. In essence, it helped teachers think more clearly about the structure and nature of knowledge. The new taxonomy helped teachers see how complex knowledge really is.

Affective Domain



The affective model actually came as a second handbook, and was an extension of Bloom's original work. It focuses on the ways in which we handle all things related to emotions, such as feelings, values, appreciation, enthusiasm, motivations, and attitudes.

Skills in the affective domain describe the way in which people react emotionally and their ability to feel other living things' pain or joy. Affective objectives typically target the awareness and growth in attitudes, emotion, and feelings.

There are five levels in the affective domain moving through the lowest-order processes to the highest.

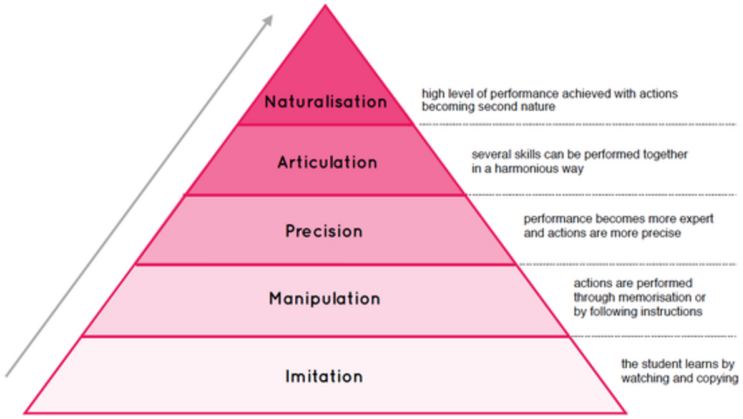
Receiving	Basic awareness, paying attention passively
Responding	Active participation, reacting to stimuli
Valuing	Associating some value to an object, a piece of information, or to the knowledge acquired
Organising	Sorting values into priorities, accommodating different values, ideas, and information into one's own schema
Characterising	Building abstract knowledge, value system is now in full effect and governs behaviour

Psychomotor Domain

The psychomotor domain of Bloom's Taxonomy refers to the ability to physically manipulate a tool or instrument. It includes physical movement, coordination, and use of the motor-skill areas. It focuses on the development of skills and the mastery of physical and manual tasks.

Mastery of these specific skills is marked by speed, precision, and distance. These psychomotor skills range from simple tasks, such as washing a car, to more complex tasks, such as operating intricate technological equipment.

Bloom and his colleagues never created subcategories for skills in the psychomotor domain, but since then other educators have created their own psychomotor taxonomies.



The first time a model for psychomotor taxonomy was published was in the year 1970, by Robert Armstrong and his colleagues. It included five levels: **imitation, manipulation, precision, articulation,** and **naturalisation.**

These tiers represent different degrees of performing skill, from exposure to mastery.

Two years later, in 1972, Anita Harrow proposed a revised version with six levels: **reflex movements, fundamental movements, perceptual abilities, physical abilities, skilled movements,** and **non-discursive communication.**

This model is concerned with developing physical fitness, dexterity, agility, and body control and focuses on varying degrees of coordination, from reflexes to highly expressive movements.

In the same year, Elizabeth Simpson proposed a taxonomy of seven levels: **perception, set, guided response, mechanism, complex overt response, adaptation,** and **origination.** Each domain consisted of keywords that helped to identify them. For example, perception, which was the ability to use sensory cues to guide motor activity, had keywords such as *detects, chooses, describes,* and *differentiates.*

Perception	Basic awareness
Set	Readiness to act; mental, physical, and emotional mindsets that determine action
Guided Response	Beginning stage of mastering a physical skill; requires trial and error
Mechanism	Intermediate stage of mastering, involves converting learned responses into habitual reactions
Complex Overt Response	Skillfully performing complex movements automatically and without hesitation
Adaptation	Modification of developed skills based on requirements
Origination	Ability to create new movements depending on the situation or problem

Conclusion

Teachers worldwide have a proper tool to guide the development of assignments, assessments, and overall curricula, only thanks to Bloom's taxonomy. The taxonomy explains that,

- Before you can understand a concept, you need to remember it.
- To apply a concept, you need first to understand it.
- To evaluate a process, you need first to analyse it.
- To create something new, you need to have completed a thorough evaluation

It guides modern education, and takes students through a process of synthesising information that allows them to think critically. Bloom's taxonomy accomplishes the seemingly daunting task of taking the important and complex topic of thinking and giving it a concrete structure.

However, even though it is still used today, this taxonomy does not come without its flaws. As mentioned before, the initial 1956 taxonomy presented learning as a static concept. Although this was ultimately addressed by the 2001 revised version that included active verbs to emphasise the dynamic nature of learning, Bloom's updated structure is still met with multiple criticisms.

Many psychologists take issue with the pyramid nature of the taxonomy. The shape creates the false impression that these cognitive steps are discrete and must be performed independently of one another. However, most tasks require several cognitive skills to work in tandem with each other. In other words, a task will not be only an analysis or a comprehension task. Rather, they occur simultaneously as opposed to sequentially.

Another problem with the taxonomy is that the sheer order of elements is inaccurate. When we learn, we don't always start with remembering and then move on to comprehension and creating something new. Instead, we mostly learn by applying and creating. The act of doing is where the learning lies, as opposed to moving through a regimented, linear process. Despite these several valid criticisms of Bloom's Taxonomy, this model is still widely used today.

References

1) Taxonomy: Definition - Wikipedia
https://en.wikipedia.org/wiki/Taxonomy

2) Bloom’s Taxonomy - Wikipedia
https://en.wikipedia.org/wiki/Bloom's_taxonomy

3) Bloom’s Taxonomy - SimplyPsychology
https://www.simplypsychology.org/blooms-taxonomy.html